

FMUSER FUQ-4653 Encoder Modulator 1 HDMI-1 SDI Input Supporting DVB-C-T-ATSC-ISDB-T RF ASI IP Output MPEG2 MPEG4 AVC H.264 Encoding Multiplexing for Broadcast Hospitality



Product Overview

The FUQ-4653 series integrates encoding, multiplexing, and modulation, featuring modular input modules for HDMI and SDI and outputs including ASI and IP. It processes signals by encoding video in MPEG2 or MPEG4 AVC H.264 and audio in formats like MPEG4-AAC and Dolby Digital AC3. With a modular drawer design and web-based NMS for control, it supports MPTS/SPTS streaming for broadcast and public venue applications.

Key feature

Offers Low Delay Encoding: With multiple delay modes for SDI and HDMI inputs.

Technical Specifications

HDMI Input	Video	Input	HDMI*1(the other is backup)	HDMI*1(the other is backup)	HDMI*1(the other is backup)
HDMI Input	Video	Encoding	MPEG2; MPEG4 AVC/H.264	MPEG2; MPEG4 AVC/H.264	MPEG2; MPEG4 AVC/H.264
HDMI Input	Video	Bitrate	1-19.5Mbps	1-19.5Mbps	1-19.5Mbps

HDMI Input	Video	Resolution	1920*1080_60P, 1920*1080_50P, (-for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i	1920*1080_60P, 1920*1080_50P, (-for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i	1920*1080_60P, 1920*1080_50P, (-for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i
HDMI Input	Video	Low Delay	Normal, Mode 1, Mode 2	Normal, Mode 1, Mode 2	Normal, Mode 1, Mode 2
HDMI Input	Video	Chroma	4:2:0	4:2:0	4:2:0
HDMI Input	Video	Aspect Ratio	16:9,4:3	16:9,4:3	16:9,4:3
HDMI Input	Audio	Encoding	MPEG1 Layer II;MPEG 2-AAC; MPEG 4-AAC and Dolby Digital AC3 2.0	MPEG1 Layer II;MPEG 2-AAC; MPEG 4-AAC and Dolby Digital AC3 2.0	MPEG1 Layer II;MPEG 2-AAC; MPEG 4-AAC and Dolby Digital AC3 2.0
HDMI Input	Audio	Sample rate	48KHz	48KHz	48KHz
HDMI Input	Audio	Bitrate	64/96/128/ 192/256/320kbps	64/96/128/ 192/256/320kbps	64/96/128/ 192/256/320kbps
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Video (HDMI)	Encoding	MPEG2; MPEG4 AVC/H.264	MPEG2; MPEG4 AVC/H.264	MPEG2; MPEG4 AVC/H.264
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Video (HDMI)	Input	HDMI*1	HDMI*1	HDMI*1
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Video (HDMI)	Bitrate	1-19.5Mbps	1-19.5Mbps	1-19.5Mbps

HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Video (HDMI)	Resolution	1920*1080_60P, 1920*1080_50P, -for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i	1920*1080_60P, 1920*1080_50P, -for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i	1920*1080_60P, 1920*1080_50P, -for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Video (HDMI)	Low Delay	Normal, Mode 1, Mode 2	Normal, Mode 1, Mode 2	Normal, Mode 1, Mode 2
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Video (HDMI)	Chroma	4:2:0	4:2:0	4:2:0
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Video (HDMI)	Aspect Ratio	16:9,4:3	16:9,4:3	16:9,4:3
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Audio (HDMI)	Encoding	MPEG1 Layer II ,MPEG2-AAC, MPEG4-AAC and Dolby Digital AC3 2.0	MPEG1 Layer II ,MPEG2-AAC, MPEG4-AAC and Dolby Digital AC3 2.0	MPEG1 Layer II ,MPEG2-AAC, MPEG4-AAC and Dolby Digital AC3 2.0
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Audio (HDMI)	Input	HDMI*1	HDMI*1	HDMI*1
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Audio (HDMI)	Sample rate	48KHz	48KHz	48KHz
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Audio (HDMI)	Bitrate	64/96/128/ 192/256/320kbps	64/96/128/ 192/256/320kbps	64/96/128/ 192/256/320kbps

HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Video (YpbPr/ CVBS)	Encoding	MPEG2; MPEG4 AVC/H.264	MPEG2; MPEG4 AVC/H.264	MPEG2; MPEG4 AVC/H.264
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Video (YpbPr/ CVBS)	Input	YpbPr*1 / CVBS *1	YpbPr*1 / CVBS *1	YpbPr*1 / CVBS *1
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Video (YpbPr/ CVBS)	Bitrate	1-19.5Mbps	1-19.5Mbps	1-19.5Mbps
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Video (YpbPr/ CVBS)	Resolution	CVBS: 720x576_50i (PAL); 720x4 80_60i (NTSC) YpbPr: 1920* 1080_60i, 19 20*1080_50i; 1280*720_60 p, 1280*720_ 50P	CVBS: 720x576_50i (PAL); 720x4 80_60i (NTSC) YpbPr: 1920* 1080_60i, 19 20*1080_50i; 1280*720_60 p, 1280*720_ 50P	CVBS: 720x576_50i (PAL); 720x4 80_60i (NTSC) YpbPr: 1920* 1080_60i, 19 20*1080_50i; 1280*720_60 p, 1280*720_ 50P
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Video (YpbPr/ CVBS)	Low Delay	Normal, Mode 1, Mode 2	Normal, Mode 1, Mode 2	Normal, Mode 1, Mode 2
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Video (YpbPr/ CVBS)	Chroma	4:2:0	4:2:0	4:2:0
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Video (YpbPr/ CVBS)	Aspect Ratio	16:9, 4:3	16:9, 4:3	16:9, 4:3
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Audio (YpbPr/ CVBS)	Encoding	MPEG1 Layer II; MPEG2-AAC; MPEG4-AAC and Dolby Digital AC3 2.0	MPEG1 Layer II; MPEG2-AAC; MPEG4-AAC and Dolby Digital AC3 2.0	MPEG1 Layer II; MPEG2-AAC; MPEG4-AAC and Dolby Digital AC3 2.0

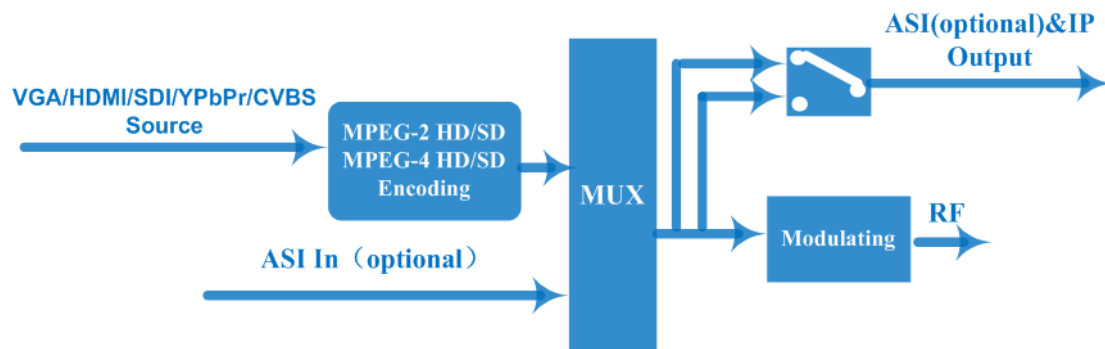
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Audio (YpbPr/ CVBS)	Interface	1*Stereo/2*mono	1*Stereo/2*mono	1*Stereo/2*mono
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Audio (YpbPr/ CVBS)	Sample rate	48KHz	48KHz	48KHz
HDMI/YPbPr/ CVBS 3-in-1 Input (under developing)	Audio (YpbPr/ CVBS)	Bit rate	64/96/128/ 192/256/320kbp s	64/96/128/ 192/256/320kbp s	64/96/128/ 192/256/320kbp s
Modulator Section	DVB-T (Optional)	Standard	DVB-T COFDM	DVB-T COFDM	DVB-T COFDM
Modulator Section	DVB-T (Optional)	Bandwidth	6M, 7M, 8M	6M, 7M, 8M	6M, 7M, 8M
Modulator Section	DVB-T (Optional)	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulator Section	DVB-T (Optional)	Code rate	1/2, 2/3, 3/4, 5/6, 7/8.	1/2, 2/3, 3/4, 5/6, 7/8.	1/2, 2/3, 3/4, 5/6, 7/8.
Modulator Section	DVB-T (Optional)	Guard Interval	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4
Modulator Section	DVB-T (Optional)	Transmission Mode	2K, 4K, 8K	2K, 4K, 8K	2K, 4K, 8K
Modulator Section	DVB-T (Optional)	MER	Appx 40dB	Appx 40dB	Appx 40dB
Modulator Section	DVB-T (Optional)	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulator Section	DVB-T (Optional)	RF Out	4*DVB-T carriers combined output	4*DVB-T carriers combined output	4*DVB-T carriers combined output
Modulator Section	DVB-T (Optional)	RF output level	-20 ~ -3dbm, 0.1db step	-20 ~ -3dbm, 0.1db step	-20 ~ -3dbm, 0.1db step
Modulator Section	DVB-C (Optional)	Standard	J.83A (DVB-C), J.83B, J.83C	J.83A (DVB-C), J.83B, J.83C	J.83A (DVB-C), J.83B, J.83C

Modulator Section	DVB-C (Optional)	MER	Appx 40dB	Appx 40dB	Appx 40dB
Modulator Section	DVB-C (Optional)	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulator Section	DVB-C (Optional)	RF output level	-20~ +3dbm, 0.1db step	-20~ +3dbm, 0.1db step	-20~ +3dbm, 0.1db step
Modulator Section	DVB-C (Optional)	Symbol rate	3.000~9.000 Msps adjustable	3.000~9.000 Msps adjustable	3.000~9.000 Msps adjustable
Modulator Section	DVB-C (Optional)	RF Out	4*DVB-C carriers combined output	4*DVB-C carriers combined output	4*DVB-C carriers combined output
Modulator Section	DVB-C (Optional)		J.83A	J.83B	J.83C
Modulator Section	DVB-C (Optional)	Constellation	16/32/64/128/256QAM	64/ 256 QAM	64/ 256 QAM
Modulator Section	DVB-C (Optional)	Bandwidth	8M	6M	6M
Modulator Section	ATSC (Optional)	Standard	ATSC A/53	ATSC A/53	ATSC A/53
Modulator Section	ATSC (Optional)	RF frequency	50~960MHz, 1KHz step.	50~960MHz, 1KHz step.	50~960MHz, 1KHz step.
Modulator Section	ATSC (Optional)	RF Out	4*ATSC carriers combined output	4*ATSC carriers combined output	4*ATSC carriers combined output
Modulator Section	ATSC (Optional)	RF output level	-20~ +3dbm, 0.1db step	-20~ +3dbm, 0.1db step	-20~ +3dbm, 0.1db step
Modulator Section	ATSC (Optional)	Constellation	8VSB	8VSB	8VSB
Modulator Section	ISDB-T (Optional) (under developing)	Standard	ARIB STD-B31	ARIB STD-B31	ARIB STD-B31
Modulator Section	ISDB-T (Optional) (under developing)	Bandwidth	6M	6M	6M

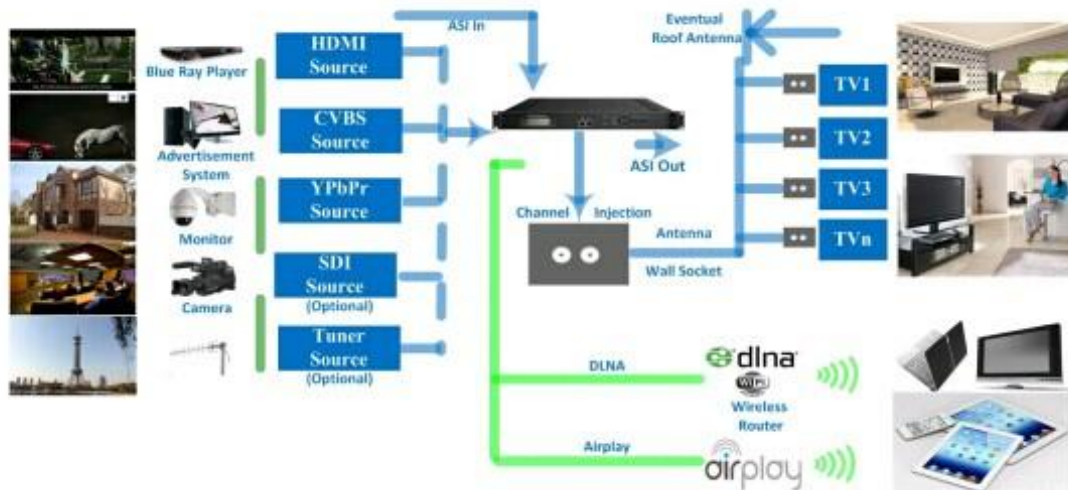
Modulator Section	ISDB-T (Optional) (under developing)	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulator Section	ISDB-T (Optional) (under developing)	Guard Interval	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4
Modulator Section	ISDB-T (Optional) (under developing)	Transmission Mode	2K, 4K, 8K	2K, 4K, 8K	2K, 4K, 8K
Modulator Section	ISDB-T (Optional) (under developing)	MER	Appx 40dB	Appx 40dB	Appx 40dB
Modulator Section	ISDB-T (Optional) (under developing)	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulator Section	ISDB-T (Optional) (under developing)	RF Out	1*ISDBT	1*ISDBT	1*ISDBT
Modulator Section	ISDB-T (Optional) (under developing)	RF output level	-20~ +3dbm, 0.1db step	-20~ +3dbm, 0.1db step	-20~ +3dbm, 0.1db step
General	System	Local interface	LCD + control buttons	LCD + control buttons	LCD + control buttons
General	System	Remote management	Web NMS	Web NMS	Web NMS
General	System	Stream Out	2 ASI out (BNC type, same one TS, mirror as MPTS or SPTS)	2 ASI out (BNC type, same one TS, mirror as MPTS or SPTS)	2 ASI out (BNC type, same one TS, mirror as MPTS or SPTS)

General	System	Stream Out	DVB-C/-T/AT SC: IP (4 MPTS & 2 SPTS) out over UDP, RTP/RTSP (4 RF carriers out) ISDBT: IP (1 MPTS & 2 SPTS) out over UDP, RTP/RTSP (1 RF carrier out)	DVB-C/-T/AT SC: IP (4 MPTS & 2 SPTS) out over UDP, RTP/RTSP (4 RF carriers out) ISDBT: IP (1 MPTS & 2 SPTS) out over UDP, RTP/RTSP (1 RF carrier out)	DVB-C/-T/AT SC: IP (4 MPTS & 2 SPTS) out over UDP, RTP/RTSP (4 RF carriers out) ISDBT: IP (1 MPTS & 2 SPTS) out over UDP, RTP/RTSP (1 RF carrier out)
General	System	DATA Port	1000M	1000M	1000M
General	System	NMS interface	RJ45, 100M	RJ45, 100M	RJ45, 100M
General	System	Language	English	English	English
General	Physical Specification	Power supply	100~240VAC, 50/60Hz	100~240VAC, 50/60Hz	100~240VAC, 50/60Hz
General	Physical Specification	Dimensions	482*242*44m m	482*242*44m m	482*242*44m m
General	Physical Specification	Operation temp	0~45■	0~45■	0~45■

Principle Chart



Typical Applications



About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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